



L.J. INSTITUTES OF COMPUTER APPLICATIONS									
Programme	Master of Computer Applications					Branch/Spec.	Computer Applications		
Semester	II					Effective from Academic Year		2025-26	
Subject Category	Minor					Effective for the batch Admitted in		June 2025	
Subject Code	240110204		Subject Name		Software Engineering and Testing (SE)				
Teaching scheme					Examination scheme (Marks)				
(Lecture hours Per week)	Credit	L	T	P	Total	E	CEC	V	Total
4	4	3	1	0	48	50	50	0	100
University Exam Evaluation Scheme									
Internal Evaluation					External Evaluation				
Mid Sem (30 Marks)		20			Theory Exam		50		
Project of Software Testing		10							
Attendance		10			Practical Exam		NA		
Theory Assignment		10							
Total Marks		50			Total Marks		50		
Pre-requisites:									
Systems & Object-Oriented Design Methodologies									
Program Learning Outcome:									
Name of PO		Description							
PO1		Foundational & Theoretical Mastery							
PO2		Software Design & Development Expertise							
PO3		Problem-Solving & Analytical Ability							
PO4		Proficiency in Emerging Technologies							
PO5		Information Systems & Management Perspective							
PO6		Professional & Ethical Responsibility							
PO7		Communication & Teamwork							
PO8		Lifelong Learning & Career Readiness							
Course Learning Outcome:									
Name of CO		Description							
CO1		Understand software engineering fundamentals, SDLC, process models, and the role of a software engineer.							
CO2		Apply Agile methodology, Scrum, and Agile principles for iterative software							



	development.
CO3	Analyze requirements, create SRS, and design software using UML diagrams and architecture principles.
CO4	Evaluate software quality, apply SQA standards, and perform software maintenance.
CO5	Apply software testing techniques and automation tools like Selenium to ensure reliable software.

Mapping of CO and PO:

Course Learning Objectives	Program Outcomes (POs)							
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	2	1	1	1	2	1	1
CO2	2	2	3	2	2	1	1	2
CO3	2	3	3	2	2	1	1	2
CO4	2	2	2	3	3	2	1	2
CO5	1	2	3	2	3	1	1	2

Content:

Unit	Content	Weightage (%)	No of Hrs.	CO
Unit I	Foundations of Software Engineering: Introduction to Software, Software Engineering Overview, Role of Software Engineer, Characteristics of Good Software, Software Crisis, Software Development Life Cycle (SDLC), Software Process, Software Process Models(Waterfall, Incremental, Prototyping, Spiral, Agile Model) Introduction to Agile Methodology: Agility and Agile Process model, Extreme Programming, Agile Manifesto and Values, Agile Principles: Overview of the 12 principles of Agile software development, Scrum and Self-Organizing Teams.	20%	10	CO1
Unit II	Requirement Engineering & Design Concept: Understanding the Requirements, Requirement Modeling, Requirement Specification, and Requirements Engineering: Seven distinct tasks of requirement engineering, Gathering functional/non-functional requirements, Create SRS. Software Design and Architecture: Design concepts and design principle, Architectural design, design UML diagrams: (Use case, class, sequence, activity diagram)	20%	10	CO2
Unit III	Software Quality Assurance and Software Maintenance:	20%	10	CO3



	Quality Concept and Software Quality Assurance, Software Reviews (Formal, Technical Review), Software Reliability. The Quality Standards: ISO 9000, CMM, Six Sigma for SE,SQA plan Software Maintenance: Types of Software Maintenance, Re-Engineering, Reverse Engineering, Forward Engineering			
Unit IV	Software Testing Fundamentals: What is software testing, Why it is important? Verification, Validation, Qualification, STLC, Testing Basics: Levels: Unit, Integration, System, Acceptance, Types: Manual, Automation, functional, non-functional, Techniques: Black-box, White-box, Grey-box, Test Case Design: Techniques: Boundary Value Analysis, Equivalence Partitioning, Writing effective test plans and test cases	20%	10	CO4
Unit V	Advanced Testing and Automation Testing Tool (Selenium) Introduction to automation frameworks, How automation frameworks work? Types of automation frameworks, Where are automation frameworks used, Introduction to selenium, Features of selenium, Selenium limitations, Installation and Setup, Selenium,WebDriver Commands, Browser Commands, Navigation Commands, WebElement Commands, FindElement and FindElements Command CheckBox & Radio Button Operations, DropDown & Multiple Select Operations, Handle Dynamic WebTables in Selenium Webdriver, Locating web elements by Id, ClassName, LinkText, PartialLinkText, Name, TagName, CssSelector , XPath, Inspecting elements in web browsers.	20%	8	CO5

Reference Books:

1	Software Engineering: Roger S. Pressman & Bruce R. Maxim – <i>Software Engineering: A Practitioner's Approach</i>
2	Software Engineering: Ian Sommerville – <i>Software Engineering</i>
3	UML & Design: Craig Larman – <i>Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development</i>
4	UML & Design: Martin Fowler – <i>UML Distilled: A Brief Guide to the Standard Object Modeling Language</i>
5	Software Testing: Glenford J. Myers, Corey Sandler & Tom Badgett – <i>The Art of Software Testing</i>

Web Reference:

1	Software Engineering Institute (SEI), Carnegie Mellon University: https://www.sei.cmu.edu
2	Agile Alliance (Agile & Scrum concepts): https://www.agilealliance.org
3	Selenium Official Documentation (Automation Testing): https://www.selenium.dev/documentation

Question Paper Scheme:

	University Examination Duration: 2 Hours 30 minutes.
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SECTION-I

Q-1: 10 Marks

Q-2: 10 Marks

Q-3: 10 Marks

Q.4: 10 Marks

Q.5: 10 Marks